

NAME

s390_pci_mmio_write, s390_pci_mmio_read – transfer data to/from PCI MMIO memory page

LIBRARY

Standard C library (*libc*, *-lc*)

SYNOPSIS

```
#include <sys/syscall.h>    /* Definition of SYS_* constants */
#include <unistd.h>

int syscall(SYS_s390_pci_mmio_write, unsigned long mmio_addr,
            const void user_buffer[.length], size_t length);
int syscall(SYS_s390_pci_mmio_read, unsigned long mmio_addr,
            void user_buffer[.length], size_t length);
```

Note: glibc provides no wrappers for these system calls, necessitating the use of `syscall(2)`.

DESCRIPTION

The `s390_pci_mmio_write()` system call writes *length* bytes of data from the user-space buffer *user_buffer* to the PCI MMIO memory location specified by *mmio_addr*. The `s390_pci_mmio_read()` system call reads *length* bytes of data from the PCI MMIO memory location specified by *mmio_addr* to the user-space buffer *user_buffer*.

These system calls must be used instead of the simple assignment or data-transfer operations that are used to access the PCI MMIO memory areas mapped to user space on the Linux System z platform. The address specified by *mmio_addr* must belong to a PCI MMIO memory page mapping in the caller's address space, and the data being written or read must not cross a page boundary. The *length* value cannot be greater than the system page size.

RETURN VALUE

On success, `s390_pci_mmio_write()` and `s390_pci_mmio_read()` return 0. On failure, `-1` is returned and *errno* is set to indicate the error.

ERRORS**EFAULT**

The address in *mmio_addr* is invalid.

EFAULT

user_buffer does not point to a valid location in the caller's address space.

EINVAL

Invalid *length* argument.

ENODEV

PCI support is not enabled.

ENOMEM

Insufficient memory.

STANDARDS

Linux on s390.

HISTORY

Linux 3.19. System z EC12.

SEE ALSO

`syscall(2)`